

Road Infrastructure, Foreign Direct Investment, and Economic Growth in China

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November 12, 2025

Declaration of Competing Interest: The authors declare no competing financial or personal interests that could have influenced the work reported in this paper.

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Abstract

Using a panel of Chinese cities over the period 1999 - 2018, we examine the determinants of economic growth, focusing on the role of foreign direct investment (FDI) and road infrastructure. Consistent with the predictions of a human capital-augmented Solow model, we find that FDI has a positive effect on the per capita GDP growth rate and this effect is intensified by the road infrastructure of the city. The latter suggests one way that road infrastructure contributes to growth is to serve as a facilitator for technology transfers stemming from FDI. The findings suggest the complementary effect of FDI on road infrastructure is stronger for technology-intensive cities, coastal regions, and first-tier cities. The results are robust to alternative model specifications and estimation methods.

Keywords: Road Infrastructure, Economic Growth, FDI, Development

JEL Codes: F21, O18, O53, R40,

1 Introduction

Transportation infrastructure plays a crucial role in economic development, especially for developing countries, by reducing transport costs, improving regional connectivity, and facilitating trade and access to services. China provides a prominent example, having invested heavily in its road network over the past few decades. The country now boasts one of the world's largest highway systems (the National Trunk Highway System), with an expressway length of about 161,000 kilometers as of 2020 – the longest in the world. These extensive highway investments reflect China's strategy of leveraging infrastructure development to support rapid growth in output and trade. Indeed, the expansion of China's highways has been extraordinary. For instance, since 1990 the growth rate of highway length has far outpaced GDP growth (Figure 1), underscoring the idea that initial access to markets and ideas is a prerequisite for broad-based development (Banerjee et al., 2020). Historically, countries undergo phases of transportation development – from establishing basic connectivity to rapidly expanding networks to upgrading services – and China has been swiftly moving through these stages.

In the early reform era, China's transportation infrastructure development lagged behind the pace of economic growth, creating bottlenecks that constrained further progress. Inadequate transport links, especially in interior regions, were a significant factor behind regional growth disparities (Démurger, 2001). Recognizing this, the Chinese government undertook massive highway expansion: since 1978, the total length of highways has increased more than thirty-fold, accompanied by a dramatic rise in freight volumes and connectivity. By 2020, road transport accounted for nearly 74% of China's total freight tonnage, reflecting the pivotal economic role of improved infrastructure (National Bureau of Statistics of China, 2021). Given these trends, this paper focuses on road infrastructure - particularly highways - and its relationship with economic growth, rather than examining all types of infrastructure.

Parallel to infrastructure development, China's economy experienced unprecedented growth after 1978, averaging about 10% annual GDP growth for three decades (Holz, 2008; Whalley and Xin, 2010). This remarkable performance was fueled in large part by the surge in international trade and inflows of foreign direct investment (FDI) (Zhao and Du, 2007; Gunby et al., 2017). FDI has played a crucial role by bringing in capital and advanced technology, which often have a larger impact on productivity than domestic investment. Initially, the bulk of FDI was concentrated in the eastern coastal provinces - over 80% of FDI during 1980–2000 went to the coast - due in part to those regions' better market access and infrastructure, as well as policy incentives (Fu, 2004). In contrast, the central and western regions attracted comparatively less FDI, hampered by underdeveloped transport networks and less favorable geographic location. Empirical evidence confirms that areas with good infrastructure tend to attract more FDI; for example, Cheng and

Kwan (2000a) find that large regional markets and robust infrastructure had a significant positive effect on FDI distribution in China. Over time, as development policies shifted westward and highway connectivity improved across the interior, foreign capital began flowing increasingly into central and western areas. Improved roads have thus expanded the geographic reach of foreign investment, potentially reinforcing growth in regions that were previously left behind (see Figure 2).

FDI's contribution to economic growth has been widely studied. Beyond providing capital, FDI facilitates technology transfer and productivity spillovers to domestic firms. Figure 3 depicts the fluctuations in both FDI and real GDP growth since 1990. We see that the progression of economic development in China is evident through a significant increase in FDI inflows into the Chinese market. FDI plays a crucial role in facilitating technology transfer (Blomström and Wang, 1989; Iwasaki and Tokunaga, 2016). When compared to domestic investment, FDI makes a relatively more substantial contribution to economic growth owing to its utilization of advanced and higher-level technologies (Borensztein et al., 1998). However, the magnitude of these benefits depends on the host economy's absorptive capacity - factors such as human capital, openness, and innovation capability determine how effectively foreign technologies are assimilated. For example, Borensztein et al. (1998) argue that FDI's positive growth impact materializes only when the host country has a minimum threshold of human capital. Thus, while FDI can be a powerful engine of growth through competition and innovation spillovers, its impact is conditional on local circumstances (e.g., education levels, institutional quality) that enable technology absorption (Liu et al., 2001; Fu, 2008; Su and Liu, 2016).

Another strand of literature examines how improvements in transportation infrastructure affect economic performance. Numerous studies find that better roads and highways enhance productivity and market integration, often indirectly promoting growth through trade expansion. For instance, major highway investments have been shown to facilitate greater trade flows and regional specialization rather than directly increasing local GDP growth rates (Duranton et al., 2014; Redding and Turner, 2015). In the United States, the expansion of the interstate highway system stimulated economic activity in connected counties and influenced the spatial distribution of industries (Chandra and Thompson, 2000; Michaels, 2008). Similarly, in India, the Golden Quadrilateral national highway project significantly reduced travel times and inventory costs for firms, thereby allowing manufacturers to serve larger markets more efficiently (Datta, 2012). Cross-country surveys also underscore that transportation infrastructure plays a fundamental role in economic organization by connecting markets and reducing trade costs (Redding and Turner, 2015).

Within China, infrastructure investment has proven crucial for regional development and integration. Démurger (2001) finds that transport infrastructure was a key differentiating factor

in explaining the growth gap among Chinese provinces - regions with denser highway and rail networks grew significantly faster, contributing to reduced regional disparities. In rural China, [Shenggen and Zhang \(2004\)](#) show that government spending on roads (along with irrigation) significantly boosted agricultural productivity and rural incomes. Recent research also confirms positive impacts of transportation improvements on China's economy. For example, [Roberts et al. \(2012\)](#) and [Banerjee et al. \(2020\)](#) find that the construction of China's national expressway network had a positive causal effect on per capita income levels across regions (even if it did not significantly change short-run GDP growth rates). Investments in other transport modes, such as high-speed rail, have likewise been linked to higher local economic output (e.g., [Ke et al. \(2017\)](#) on the effects of China's high-speed rail). That said, some studies highlight that infrastructure benefits can be unevenly distributed: for instance, [Faber \(2014\)](#) exploits the rollout of the National Trunk Highway System (NTHS) as a natural experiment and finds that while the new highways improved market accessibility, they also led to reduced GDP growth in certain peripheral counties due to economic activity becoming concentrated in major city hubs. On balance, however, the literature suggests that robust road infrastructure is an essential ingredient for enabling trade, reducing geographic isolation, and fostering productivity growth - all of which lay the groundwork for higher long-term growth ([Demurger, 2001](#); [Shenggen and Zhang, 2004](#)).

A related body of work investigates how geography and infrastructure influence the location of FDI. In general, foreign investors gravitate toward regions with large markets, good infrastructure, and conducive policies ([Cheng and Kwan, 2000b](#); [Halaszovich and Kinra, 2020](#)). Studies of U.S. manufacturing FDI, for example, find that states with more extensive transportation networks attract significantly more foreign investment ([Coughlin et al., 1991](#)). In the Chinese context, [Cheng and Kwan \(2000b\)](#) similarly conclude that provinces with better infrastructure (and access to ports) received higher FDI inflows, controlling for other factors. The reason is straightforward: by lowering transportation and transaction costs, infrastructure effectively expands the profitable "reach" of foreign enterprises. Other research emphasizes the role of distance and trade costs in FDI location decisions - for instance, if overseas markets are very distant, multinationals may choose exporting over local investment, unless efficient transport infrastructure mitigates those distance-related costs. Consistent with this, [Halaszovich and Kinra \(2020\)](#) provide evidence that a country's transportation systems moderate the adverse effects of geographical distance on FDI and trade performance.

Despite extensive research on FDI-led growth and on infrastructure-led growth separately, relatively few studies have analyzed how these two factors interact with each other to influence economic growth. Most of the existing literature tends to examine either how infrastructure affects FDI flows or how each factor independently impacts growth, without explicitly considering their joint effects. Notably, to the best of our knowledge, no prior study has directly explored

the growth consequences of the interaction between transportation infrastructure and FDI. This paper aims to fill this gap in the literature. We investigate whether road infrastructure and FDI have a complementary effect on regional economic growth in China. In other words, do regions with both strong road networks and high FDI inflows experience faster growth than would be predicted by each factor alone? We hypothesize that the interaction of FDI and highways exerts a positive influence on local economic growth. The intuition is that improved highways may amplify the growth benefits of FDI by better connecting firms to markets and suppliers, while incoming FDI can help regions more fully capitalize on infrastructure by bringing in investment, jobs, and technology. This mutually reinforcing dynamic between FDI and road infrastructure could be an important driver of regional growth differences in China.

Descriptive evidence is consistent with this idea of synergy. Over the past two decades, FDI inflows in China have gradually spread inland from the coast, while highway infrastructure has simultaneously expanded into the interior. The spatial distribution of FDI in 2000 versus 2018 (Figures 4 and 5) shows that many interior cities with minimal FDI in 2000 had become significant FDI recipients by 2018. During the same period, China's highway network grew dramatically, connecting virtually all prefecture-level cities by 2020 (Figures 6 and 7). These parallel trends suggest that roads and FDI have been advancing in tandem. It stands to reason that this co-movement - better infrastructure attracting more FDI, and FDI in turn leveraging the infrastructure - has contributed to faster growth in those regions that benefited from both improvements. This study formally tests and confirms that notion within a growth regression framework.

This paper makes several contributions to the literature. First, we introduce a panel data analysis based on an augmented Solow model to examine the determinants of economic growth across Chinese cities. By focusing on growth along the transition path to each city's steady-state income (rather than just long-run income levels), we ground our empirical specification in a well-established theoretical framework. This approach provides clear guidance on including fundamental variables (such as saving/investment rates and population growth) and helps avoid ad-hoc model specifications. Consistent with the Solow model's predictions, we find that a city's per capita GDP growth rate is negatively related to its population growth and positively related to its physical and human capital investment rates. Within this framework, we show that road infrastructure emerges as a significant factor positively influencing the transitional growth rate of per capita income.

Second, unlike many studies that use a crude proxy for infrastructure (such as total road length or paved road area), we compile a more nuanced measure that distinguishes between different types of roads. In particular, we focus on inter-city highways and expressways - the arterial roads that connect economic centers - as opposed to local urban streets. This distinction is important because major highways are the roads most likely to reduce trade costs and integrate markets

across regions. Our analysis reveals that the highway system indeed promotes economic growth and, importantly, enhances the effect of FDI on growth. In other words, we find evidence of complementarity: FDI has a larger growth impact in locations with extensive highway networks than in those with poor road connectivity. By contrast, we find that paved urban roads (municipal streets within city limits) do not always significantly intensify FDI's effect on growth. This result suggests it is the broad connectivity provided by highways, rather than local road density per se, that matters more for leveraging FDI. Our refined infrastructure measure thus contributes to the literature by highlighting the differential growth effects of trunk highways versus local roads. It also addresses a measurement gap noted in prior research, which often emphasized the need for better infrastructure metrics when examining productivity (e.g., the “productivity puzzle” around public capital noted by [Holts-Eakin \(1994\)](#), where controlling for regional factors showed much weaker infrastructure effects.

Third, we provide new evidence on the heterogeneity of the FDI–infrastructure growth nexus across regions with different technological profiles. We find that the complementary effect of FDI and highways on growth is especially pronounced in high-tech regions (areas with a greater concentration of technology-intensive industries). In these regions, foreign investments likely bring frontier technologies that, when combined with good infrastructure, yield substantial productivity gains. In contrast, in more labor-intensive (low-tech) regions, we do not observe that road infrastructure has a positive direct impact on growth, and there is little to no additional boost from FDI–road interaction. Moreover, consistent with our earlier finding, paved local roads show significant complementary effect with FDI in only some regions. These heterogeneous results enrich our understanding of how local conditions (industrial structure, tech intensity) mediate the benefits of infrastructure and FDI. They suggest that policymakers in high-tech regions might gain especially high returns from coordinating infrastructure upgrades with efforts to attract FDI, whereas in lower-tech regions the gains from FDI–infrastructure synergy may be more limited.

Finally, our study carefully addresses potential endogeneity concerns. It is possible that causality runs in the reverse direction - i.e., richer or faster-growing cities invest more in roads and attract more FDI, which could bias the results of simple growth regressions. This issue of reverse causality and omitted variable bias in growth–infrastructure studies has been highlighted by prior research ([Holts-Eakin, 1994](#)). We tackle this challenge in two ways. First, our panel fixed-effects model accounts for unobserved city-specific factors (such as geography or historical institutions) that could affect both growth and infrastructure investment, thereby mitigating omitted variable bias. Second, we implement an instrumental variable (IV) approach following the strategy of [Feng and Wu \(2018\)](#): we use lagged infrastructure development as an instrument to isolate plausibly exogenous variation in current road infrastructure. The IV regressions yield consistent estimates that reinforce our main findings. In fact, the estimated FDI–road comple-

mentarity effect on growth is even stronger when using the IV, suggesting that our baseline may understate the true effect. These robustness checks give us confidence that the positive interaction effect we document is not driven by spurious correlation. Our attention to identification echoes broader concerns in the growth literature (e.g. [Hauk Jr \(2017\)](#) on endogeneity bias) and ensures that our contributions are built on a solid empirical foundation.

To the best of our knowledge, this paper is the first to examine the interactive impact of road infrastructure and FDI on regional economic growth in China. We find that highways and FDI reinforce each other in promoting growth: the presence of a strong road network significantly increases the growth dividends from FDI. We also contribute methodologically by using an augmented Solow framework with improved infrastructure measures and by addressing endogeneity rigorously. These findings have direct relevance for policy, suggesting that investments in transportation infrastructure and efforts to attract FDI can yield greater returns if pursued in tandem. For developing countries like China that seek to sustain growth and reduce regional disparities, fostering this synergy between infrastructure and foreign investment can be an effective strategy.

The remainder of the paper is structured as follows. Section 2 presents the theoretical model and empirical methodology. Section 3 describes the data and variables used in the analysis. Section 4 reports the empirical results, including baseline estimates and various robustness checks. Section 5 conducts heterogeneity analysis and Section 6 discusses policy implications. Section 7 concludes our paper.

2 Model

Building upon the foundational work of [Mankiw et al. \(1992\)](#) and extending the framework introduced by [Su and Liu \(2016\)](#), where each economy converges to its steady state over the long term, FDI plays a pivotal role as an efficiency factor that brings advanced technology to the equation. This, in turn, affects the rate at which recipient countries converge to their steady states.

In this study, we employ the extended Solow framework of [Su and Liu \(2016\)](#) to examine the interplay between regional FDI and road infrastructure. We introduce FDI and its interaction with road infrastructure to the neoclassical growth model. FDI serves as a carrier of advanced technology ([Su and Liu, 2016](#)), while road infrastructure acts as a catalyst that accelerates its adoption and diffusion. When combined, they not only enhance productivity and raise the steady-state level of income but also significantly increase the speed at which an economy converges to that steady state.

The model is based on a standard Cobb-Douglas production function for an economy. Additionally, we assume that technological progress is labor-augmenting, implying that the variable

A represents labor-augmenting technological progress, growing exogenously at a rate g . Assume labor grows exogenously at rate n , and the technology change is a function of FDI.

$$L(t) = L(0)e^{nt} \quad (1)$$

$$A(t) = A(0)e^{gt}F(t)^\lambda \cdot f(\text{Road}(t)) \quad (2)$$

There are three parts of Equation (2). The first part is $A(0)e^{gt}$, which describes the exogenous growth of technological progress. The second part is $F(t)$, representing the FDI. The third part is the $f(\text{Road}(t))$. $\text{Road}(t)$ is the level of road infrastructure. To simplify the empirical specification, we assume A is the exponential function of road infrastructure. $f(\text{Road}(t)) = \exp(\theta \text{Road}(t))$, θ is the parameter. This assumption defines that road infrastructure has generative effects, which increase income by utilizing resources more effectively and enhancing productivity (Kailthya and Kambhampati, 2018).

The construction of highways has established vital channels for economic connectivity between central cities across the country, while also linking small and medium-sized cities along these routes. This has the effect of reducing spatial distances between different regions, enhancing traffic accessibility, compressing both time and space, and transcending regional boundaries. These developments lead to decreased transportation costs, improved transaction efficiency, and, as a result, contribute to overall macroeconomic growth.

The third part of the equation operates under the assumption that FDI exerts a positive influence on technological advancements, driven by both direct and indirect effects. The indirect effect, on the other hand, arises from the spillover effect, where domestic firms absorb or acquire technology from foreign investments, consequently leading to improved efficiency and productivity. Additionally, a well-developed road infrastructure fosters collaboration between domestic firms and FDI, further amplifying efficiency gains. To measure the impact of FDI on productivity, we include road infrastructure as a component within the elasticity of FDI.

$$\lambda = \lambda_0 + \lambda_1 \ln(\text{Road}(t)) \quad (3)$$

Notably, We allow road infrastructure to affect growth through two distinct channels. First, roads directly raise productivity by reducing transport costs, which we capture through a multiplicative function $f(\text{Road})$. This effect is expected to exhibit diminishing marginal returns, consistent with the standard infrastructure literature (Chandra and Thompson, 2000; Datta, 2012). Second, roads also enhance the productivity of foreign investment: better highways allow foreign firms to access markets, suppliers, and distribution networks more effectively. To capture this, we let the elasticity of FDI with respect to productivity depend on $\ln(\text{Road})$. The logarithm-

mic form implies diminishing marginal returns to infrastructure in enhancing FDI efficiency, but because it enters as an exponent on FDI, the combined effect on growth can be magnified once a sufficient network is in place. This mechanism is consistent with evidence that infrastructure reduces trade costs and shapes firm location decisions (Redding and Turner, 2015), and that better transport networks in China facilitated both industrialization and foreign capital inflows (Cheng and Kwan, 2000a; Faber, 2014). In short, this distinction reconciles the two roles of roads: as a direct, concave productivity factor, and as a compounding force that amplifies the growth benefits of FDI.

According to the Solow-Swan model, the saving ratios are assumed to be exogenously determined by savers' preferences or government decisions. Therefore, assume that the accumulation of physical capital and human capital are as follows.

$$\dot{K} = s_k Y(t) - \delta K(t) \quad (4)$$

$$\dot{H} = s_h Y(t) - \delta H(t) \quad (5)$$

where $\dot{K} = dK/dt$, s_k and s_h are physical and human capital saving rates. δ is the depreciation rate. Assume the depreciation rates of physical and human capital are the same.

To specify the production function in terms of per effective labor, let $y(t) = Y(t)/A(t)L(t)$, $k(t) = K(t)/A(t)L(t)$, $h(t) = H(t)/A(t)L(t)$. Then combining with the equation (4) and (5), the change in stock of physical and human capital per effective unit of labor are yielded as following.

$$\dot{k} = s_k y(t) - (n + g + \delta)k(t) \quad (6)$$

$$\dot{h} = s_h y(t) - (n + g + \delta)h(t) \quad (7)$$

In the steady-state, the levels of capital per effective worker are constant. From equations (6) and (7), this assumption implies

$$k^* = \left(\frac{s_k^{1-\beta} s_h^\beta}{n + g + \delta} \right)^{\frac{1}{1-\alpha-\beta}} \quad (8)$$

$$h^* = \left(\frac{s_k^\alpha s_h^{1-\alpha}}{n + g + \delta} \right)^{\frac{1}{1-\alpha-\beta}} \quad (9)$$

Inserting the equations (8) and (9) into the production function and taking the log of both

sides. Then, the steady-state value of income per capita is:

$$\ln \left(\frac{Y(t)}{L(t)} \right) = \ln(A(t)) - \frac{\alpha + \beta}{1 - \alpha - \beta} \ln(n + g + \delta) + \frac{\alpha}{1 - \alpha - \beta} \ln(s_k) + \frac{\beta}{1 - \alpha - \beta} \ln(s_h) \quad (10)$$

Then together with the equations (2) and (3). To simplify the empirical specification, we assume the elasticity of technological progress concerning FDI is $\lambda = \lambda_0 + \lambda_1 \ln(\text{Road}(t))$. Therefore, the equation of steady-state income level per capita is a function of the population growth, the saving rate of investment in physical and human capital, FDI, and road infrastructure.

$$\begin{aligned} \ln \left(\frac{Y(t)}{L(t)} \right) &= \ln(A(0)) + gt + \lambda_0 \ln(F(t)) + \lambda_1 \ln(F(t)) \cdot \ln(\text{Road}(t)) + \theta \text{Road}(t) \\ &\quad - \frac{\alpha + \beta}{1 - \alpha - \beta} \ln(n + g + \delta) + \frac{\alpha}{1 - \alpha - \beta} \ln(s_k) \\ &\quad + \frac{\beta}{1 - \alpha - \beta} \ln(s_h) \end{aligned} \quad (11)$$

This paper analyzes the city-level economic growth; thus, it is inappropriate to discuss the steady-state. This study focuses on examining the transition dynamics of an economy toward its steady state. Following [Mankiw et al. \(1992\)](#), the economy approaches its steady-state defined by the following equation:

$$\frac{d \ln(y(t))}{dt} = \eta [\ln(y^*) - \ln(y(t))] \quad (12)$$

where $\eta = (n + g + \delta)(1 - \alpha - \beta)$, is the speed of convergence. The equation (12) is a differential equation in $\ln(y(t))$ with the solution:

$$\ln(y(t)) = (1 - e^{-\eta t}) \ln(y^*) + e^{-\eta t} \ln(y(0)) \quad (13)$$

Plugging (11) into (13) and subtracting $\ln(y(0))$ from both sides, the equation is obtained as

following.

$$\begin{aligned}
\ln\left(\frac{Y(t)}{L(t)}\right) - \ln\left(\frac{Y(0)}{L(0)}\right) &= \ln(A(0)) + gt - (1 - e^{-\eta t})\ln\left(\frac{Y(0)}{L(0)}\right) \\
&\quad - \frac{\alpha + \beta}{1 - \alpha - \beta}\ln(n + g + \delta) + \frac{\alpha}{1 - \alpha - \beta}(1 - e^{-\eta t})\ln(s_k) \\
&\quad + \frac{\beta}{1 - \alpha - \beta}(1 - e^{-\eta t})\ln(s_h) \\
&\quad + \lambda_0(1 - e^{-\eta t})\ln(F(t)) \\
&\quad + \lambda_1(1 - e^{-\eta t})\ln(F(t)) \cdot \ln(Road(t)) \\
&\quad + (1 - e^{-\eta t})\theta Road(t)
\end{aligned} \tag{14}$$

Re-writing the equation (14), the cross-section regression model is that:

$$\begin{aligned}
\ln(y_{it}) - \ln(y_{i0}) &= \phi_0 + \phi_1 \ln(y_{i0}) + \phi_2 \ln(n_i + g + \delta) + \phi_3 \ln(s_{k,i}) + \phi_4 \ln(s_{h,i}) \\
&\quad + \phi_5 \ln(F_i) + \phi_6 \ln(F_i) \cdot \ln(Road_i) + \phi_7 Road_i + \epsilon_i
\end{aligned} \tag{15}$$

In the equation (15), all the independent variables are the average value over the entire period, except the log of initial year income per capita. ϵ_i in the equation (15) is the error term including $A(0)$ and other random factors. According to equations (14) and (15), the model predicts the negative relationship between population growth rate and income per effective worker ($\phi_2 < 0$). Also, the saving rate of physical and human capital is positively related to income per effective labor ($\phi_3, \phi_4 > 0$). FDI helps increase the economic growth ($\phi_5 > 0$). The coefficient ϕ_6 is expected to be positive, indicating that road infrastructure amplifies the positive effect of FDI on economic growth, while ϕ_7 represents the direct positive impact of road infrastructure on economic growth.

Mankiw et al. (1992) assumed $A(0)$ is independent of the saving rate of physical capital, human capital and population growth rate, ignoring the country-specific effects. Instead of using the cross-sectional approach, we also specify the fixed-effects estimation, following the idea from Knight et al. (1993) and Su and Liu (2016). Then, the fixed-effects model is:

$$\begin{aligned}
\ln(y_{it}) - \ln(y_{i0}) &= \phi_0 + \phi_1 \ln(y_{i0}) + \phi_2 \ln(n_{it} + g + \delta) + \phi_3 \ln(s_{k,it}) + \phi_4 \ln(s_{h,it}) \\
&\quad + \phi_5 \ln(F_{it}) + \phi_6 \ln(F_{it}) \cdot \ln(Road_{it}) + \phi_7 Road_{it} + \mu_i + \epsilon_{it}
\end{aligned} \tag{16}$$

The panel approach uses non-overlapping intervals of five years. For the independent variables, except $\ln(y_{i0})$, the observations correspond to averages over the five-year intervals. Therefore, the t in the equation (16) indexes sub-periods.

3 Data

3.1 Data Resource

The dataset comprises annual observations collected from 265 prefecture-level cities in China spanning from 1999 to 2018. The primary data sources include various editions of the China City Statistical Yearbooks (1999-2018) and the China Statistical Yearbooks for Regional Economy (2000-2014). However, it's important to note that some missing values are identified in the compiled dataset. To mitigate the potential bias introduced by missing data, additional information is gathered from the Statistical Yearbooks of specific cities and manually collated from local statistical communiques for the respective cities.

The China City Statistical Yearbook is an annual statistical publication sponsored by the National Bureau of Statistics of China, providing comprehensive insights into the economic and social development of Chinese cities. It's important to note that the study region excludes Hong Kong, Macao, and Taiwan Province. When referring to the whole city in the yearbook tables, it encompasses all administrative areas within the city, including urban areas, counties, and rural areas.

3.2 Data Summary and Measurement

China's Highway System - The development of China's highways can be broadly categorized into three stages: 1. *The initial stage of steady development (1978-1997)*. 2. *The stage of rapid and leapfrog development (1998-2015)*. 3. *The stage of comprehensive and high-quality development (post-2016)*. Since 1998, during the Asian financial crisis, China implemented a proactive fiscal policy, which included substantial investments in infrastructure construction. This policy shift led to a surge in bank loans directed towards expressway construction, thereby driving the rapid expansion of expressways across China. Simultaneously, local governments actively released detailed road-related data, facilitating the collection of information and consistent variables related to road infrastructure. Additionally, it's worth noting that more convergence studies are available for the early period. Hence, we have chosen 1999 as the starting point for my analysis.

The road data used in this paper encompasses roads designed for connecting various areas, including rural areas, urban areas, and cities, and conforms to national technical standards with proper authorities' approvals. This data serves as a crucial indicator, reflecting the scale of highway construction development and acts as the primary data source for calculating metrics such as transportation network density.¹

¹This data pertains to the actual length of roads at the conclusion of the reporting period. The statistical scope encompasses public roads suitable for automobile travel that connect cities, cities to rural areas, and villages. This

The road data presented in Statistical Yearbooks is calculated in terms of kilometres. However, solely relying on kilometres as a measure to gauge the investment and construction scale of highway projects may not provide an accurate picture. To address this, we have employed the embodied technical change model developed by [Hulten \(1992\)](#) to estimate the capital associated with road infrastructure. This measurement captures the intuitive notion that technical progress is linked to improvements in the design of new road construction. The equation for road capital stock follows the method in [Hulten \(1992\)](#):

$$\text{Road}(t) = \sum_{\tau=0}^t \phi(\tau)(1 - \delta)^{t-\tau}l(\tau) \quad (17)$$

where l is the length of the road, δ is the rate of depreciation, which equals 0.096. ϕ denotes the effectiveness differential. Full effectiveness is defined as the state in which road infrastructure is preserved in optimal condition and operated under best-practice efficiency standards. The weighting factor is derived from the proportion of freight traffic volume attributed to the respective infrastructure network segment.

The data on the area of urban streets are obtained from the City Statistical Yearbook of China. It is important to clarify that the term urban street in this context specifically refers to smaller city streets that primarily support daily life activities, such as commuting to work, attending school, or shopping. This definition differs from the broader concept of road infrastructure discussed earlier.

Investment — In much of the earlier literature, fixed asset investment was commonly used as a direct proxy for overall investment ([Su and Liu, 2016](#)). However, in recent years, concerns have been raised regarding the reliability of fixed asset investment as a true measure of investment.² A key issue arises from the fact that, in some Chinese provinces, reported fixed asset investment has even exceeded GDP, suggesting overestimation and distortion.

An alternative measure is fixed capital formation, which reflects investment under the expenditure approach to GDP. Before 2005, fixed asset investment and fixed capital formation were nearly identical in value. Since then - especially from 2010 onward - a widening gap between the two has become evident.

Unfortunately, city-level data on fixed capital formation are unavailable. To address this, we

includes the lengths of highways that traverse urban streets, highway bridges, tunnel lengths, and ferry widths. However, it excludes urban street mileage, dead-end road mileage, agricultural (or forestry) production road mileage, as well as internal road mileage within industrial (or mining) enterprises.

²The credibility of fixed asset investment statistics has been questioned by industry researchers. In 2021, the Bureau of Statistics conducted a systematic revision of these data, eliminating certain inaccuracies and distortions. It should be noted, however, that this revision was implemented at the national level and did not extend to city-level statistics.

calculate the ratio of provincial-level fixed capital formation to fixed asset investment, which serves as an adjustment factor. Then, we apply this ratio to city-level fixed asset investment to derive a more accurate estimate of city-level investment.

Real GDP - Real GDP per capita is calculated as the gross domestic product per capita, directly sourced from Statistical Yearbooks and adjusted using the GDP deflator, with 1999 as the base year.

Human Capital - As per the approach outlined by [Su and Liu \(2016\)](#), a valid proxy for the saving rate of human capital was share of the total population enrolled in high-level education (college or university).

FDI - The statistical yearbooks only include the new flow of FDI annually. Therefore, we follow the idea from [Zhang \(2008\)](#), which uses the perpetual inventory method to construct the suitable proxies of stock of FDI. It is the most commonly used method for domestic and foreign researchers to measure the capital stock and is pioneered by [Goldsmith \(1951\)](#). The formula is that:

$$F_{it} = (1 - \delta)F_{i,t-1} + I_{it} \quad (18)$$

where δ is the rate of replacement, and replacement refers to the maintenance and recovery of production capacity, which equals 0.096. I_{it} is the based year stock of FDI for city i . Table (1) reports the description of the variable used.

4 Result

4.1 Cross-section Regression

Table 2 presents the cross-sectional estimation results based on Equation (15), where the dependent variable is the log difference in real GDP per capita between 1999 and 2018. All explanatory variables are measured as city-level averages over the same period. The model assumes a fixed depreciation-adjusted growth rate of 0.05, following the convention of [Mankiw et al. \(1992\)](#).

Column (1) includes only the baseline Solow-type growth regressors. The coefficient on initial income is significantly negative, consistent with conditional convergence. Both physical capital ($\ln s_k$) and human capital ($\ln s_h$) have positive and statistically significant effects on growth. However, the coefficient on the population growth rate is unexpectedly positive and insignificant, diverging from theoretical predictions. Column (2) adds FDI as a proxy for foreign technology spillovers. Its coefficient is positive but not statistically significant at conventional levels. Column (3) introduces road infrastructure (measured as average stock over time), and again, the coeffi-

cient is positive but small and insignificant. Column (4) includes both FDI and road infrastructure simultaneously. While each term alone remains insignificant, column (5) adds an interaction term between FDI and infrastructure - specified as $\ln(F) \times \ln(\text{Road})$. This interaction term has a positive coefficient, suggesting a potential complementarity between FDI and infrastructure in driving regional growth. In other words, while neither FDI nor roads alone show significant independent effects in this cross-sectional setting, their joint presence appears to amplify economic performance.

Taken together, these results lend preliminary support to the hypothesis that FDI and road infrastructure reinforce each other, though the lack of significance on the individual components may reflect limitations inherent to cross-sectional models, especially the inability to fully control for time-invariant unobserved heterogeneity. Therefore, the next section turns to panel data regressions to validate the interaction effect in a more robust setting.

4.2 Panel Regression

Table 3 presents fixed-effects panel regression results over four non-overlapping five-year periods (1999–2003, 2004–2008, 2009–2013, 2014–2018). The dependent variable is the five-year log difference in GDP per capita. This structure allows us to control for unobserved, time-invariant city characteristics and examine within-city variation over time.

Column (1) presents the baseline Solow model. The results align with theoretical expectations: the initial GDP level has a negative and highly significant effect, consistent with conditional convergence. Investment in physical and human capital significantly contributes to growth, while the effective labor growth rate remains statistically insignificant. Columns (2) and (3) incorporate FDI and road infrastructure separately. Both variables exhibit positive and statistically significant effects, suggesting that FDI inflows and improvements in transport infrastructure independently support economic growth. Column (4) includes both FDI and road infrastructure simultaneously. Both remain positive and statistically significant, and the magnitudes are close to those in columns (2)–(3), suggesting their effects are not merely picking up each other. Column (5) provides the most comprehensive specification, and the interaction term - $\ln(F) \times \ln(\text{Road})$ - is significant at the 1% level. This suggests a robust and economically meaningful complementarity between foreign capital and domestic infrastructure. Specifically, road infrastructure appears to reduce logistical frictions, allowing foreign firms to operate more efficiently and enabling domestic firms to better absorb knowledge spillovers.

Compared to the cross-sectional results, the panel approach reveals more stable and significant estimates. The fixed-effects specification helps mitigate bias from unobserved heterogeneity, while the time structure captures dynamic effects of policy and investment. The adjusted R^2 in-

creases to 0.437 in the fully specified model, indicating an improved model fit.

To explore whether alternative measures of infrastructure produce similar patterns, we extend the baseline model to include urban street length as a complementary proxy for local infrastructure. Table A.1 presents the results. Columns (2) and (3) include the urban street length and its logarithm interaction with FDI. Notably, the interaction term between FDI and urban street infrastructure ($\ln(\text{FDI}) \times \ln(\text{UrbanStreet})$) is positive and statistically significant, suggesting that the complementarity result is robust to using a more localized measure of infrastructure beyond intercity roads. This supports the interpretation that infrastructure quality - whether regional or urban - enhances the economic returns to FDI. Importantly, the coefficient on $\ln(\text{FDI}) \times \ln(\text{Road})$ remains stable when urban infrastructure is added, indicating that road-level and street-level effects are not simply capturing the same variation.

4.3 Endogeneity and Mitigate Bias

A major challenge in identifying the causal effects of FDI and road infrastructure on economic growth is the potential endogeneity of these variables. Reverse causality is a key concern: cities experiencing higher economic growth may attract more FDI and public infrastructure investment, rather than the other way around. Foreign investors often target regions with stronger economic performance, while local governments may respond to growth by expanding infrastructure. To address this issue, we employ two identification strategies.

The first strategy uses the initial-period values of FDI and road infrastructure for each five-year panel as predetermined and plausibly exogenous. Specifically, we construct initial values for each subperiod (e.g., FDI and road levels in 1999 for the 1999–2003 panel) and use these in place of contemporaneous averages. Table 4 reports the results using this approach. Columns (1)–(4) replicate the baseline specifications, confirming the positive and statistically significant roles of FDI and road infrastructure. Column (5) introduces the interaction term between the logarithm of FDI and the logarithm of road infrastructure. The coefficient on the interaction term remains positive and statistically significant at the 1% level. This reinforces the conclusion that FDI and road infrastructure are complementary in promoting economic growth. These results suggest that using initial values reduces simultaneity bias while preserving the core empirical findings.

To further address endogeneity concerns, we implement an instrumental variable (IV) strategy. Following the literature (e.g., Holtz-Eakin, 1992; Su and Liu, 2016), we use lagged values of FDI and road infrastructure, along with their interaction term, as instruments. These lagged instruments are assumed to be correlated with current-period FDI and infrastructure but uncorrelated with the current-period residuals in the growth equation.

Table 7 presents the second-stage IV estimates. Column (1) replicates the OLS benchmark,

while Column (2) displays the IV results. The coefficient on the interaction term $\ln(\text{FDI}) \times \ln(\text{Road})$ remains positive and statistically significant at the 1% level, and its magnitude is almost the same as the OLS result. The first-stage and reduced-form regression results for this robustness check are provided in Table A.2. The first-stage regressions in Table A.2 confirm strong instrument relevance. The lagged interaction term, $\ln(\text{F})_{t-1} \times \ln(\text{Road})_{t-1}$, has a statistically significant and large coefficient in the first-stage regression (Column 4), with a corresponding F-statistic well above the conventional threshold (4174.24), alleviating concerns about weak instruments. The reduced-form results (Column 1) also show a strong positive effect of the instruments on the dependent variable. Overall, these diagnostics lend credibility to the IV identification strategy.

In sum, the findings from both OLS regression and the IV strategy support the notion that the complementary relationship between FDI and road infrastructure has a positive and significant impact on local economic growth. Even when controlling for potential omitted variables, the core conclusions of this study remain robust.

5 Heterogeneity Analysis

5.1 Fixed Estimates for Selected Sub-samples (Rank of Innovation)

While the baseline model assumes that foreign direct investment (FDI) uniformly contributes to technological progress across all cities, this assumption may not hold in practice. FDI inflows differ in their technological intensity, and cities vary in their capacity to absorb such spillovers depending on their local innovation environments. Consequently, the complementary role of road infrastructure in enhancing FDI-driven growth may be more pronounced in cities with greater innovation capacity.

Since detailed city-level data on FDI composition are unavailable, we explore this heterogeneity by dividing cities into subsamples based on two proxies for local innovation capacity: the innovation index³ and the number of patents⁴. These proxies reflect both innovation inputs and outputs, and are commonly used in empirical growth and development studies. We also examine whether urban street-level infrastructure plays a differential role across these subsamples.

Columns (1) and (2) of Table 5 report results from subsamples split by the top and bottom quartiles of the Innovation Index. The interaction term between FDI and road infrastructure,

³This index is developed in collaboration with the Enterprise Big Data Research Center of Peking University and jointly developed by the National Development Research Institute of Peking University and the Longxin Data Research Institute. It offers a set of objective indicators that gauge innovation and entrepreneurship activities at the city level in China.

⁴Patent data are obtained from the China National Intellectual Property Administration (CNIPA).

$\ln(\text{FDI}) \times \ln(\text{Road})$, is positive and statistically significant in the top-innovation cities, indicating a strong complementary relationship. This suggests that cities with high innovation capacity are better equipped to utilize road infrastructure to amplify the productivity of FDI inflows, possibly due to better institutions, absorptive capacity, or a stronger presence of innovation-intensive sectors. In contrast, in low-innovation cities, the interaction term is not significant, and road infrastructure alone has a negative and significant coefficient, implying that without sufficient local innovation capabilities, road investments may not generate meaningful synergies with FDI and may even reflect inefficient overinvestment. Columns (3) and (4) use patent count rankings as an alternative proxy for innovation. The patterns are similar: the interaction term is again positive and significant in the top-patent cities, reinforcing the idea that innovation-intensive cities experience greater gains from FDI when supported by strong infrastructure. In contrast, the interaction term becomes insignificant in the bottom-patent cities.

Incorporating urban infrastructure (see table A.3, we find that $\ln(F) \times \ln(\text{UrbanStreet})$ is also positive and significant in high-innovation cities (Columns 1 and 3), but insignificant in their low-innovation counterparts (Columns 2 and 4). This suggests that local infrastructure - both regional and urban - enhances the effectiveness of FDI only when cities possess the technological readiness to exploit these gains. In sum, infrastructure-FDI complementarities are conditional on local innovation intensity.

These findings highlight an important policy insight: the effectiveness of infrastructure investment in catalyzing FDI-driven growth is conditional on local innovation capacity. In technologically advanced cities, road infrastructure not only improves connectivity but also deepens the integration of foreign capital into the local economy, leading to stronger growth multipliers. However, in regions lacking such capacity, roads primarily serve traditional functions like reducing transport costs, and their interaction with FDI is limited. This underscores the importance of aligning infrastructure development strategies with regional innovation policies. For regions with strong R&D and innovation ecosystems, policymakers should prioritize attracting high-tech FDI and upgrading transport infrastructure in tandem. For lagging areas, complementary policies, such as human capital development, support for innovation diffusion, and public R&D investment, may be necessary to fully leverage the potential of infrastructure.

5.2 Fixed Estimates for Selected Sub-samples (Geographical or Policy conditions)

Geographical location, administrative level, and FDI policies are additional factors that influence the technical content. Coastal regions inherently possess unique geographical advantages. The presence of numerous ports has positioned many coastal cities as pioneers in attracting for-

eign investment, as illustrated in Figures 4 and 5. This section investigates whether the complementarity between road infrastructure and FDI varies by region, administrative status, or the presence of preferential FDI policies.

Columns (1) and (2) of Table 6 show the results for coastal and inland cities, respectively. In coastal regions, the interaction term between FDI and road infrastructure is positive and statistically significant at the 1% level (coefficient = 0.016), and notably larger than the corresponding coefficient in inland cities (0.007). This suggests that the complementary effect of FDI and road infrastructure is stronger in coastal areas, likely due to their better connectivity, developed logistics systems, and higher baseline of foreign investment. However, in coastal cities, road infrastructure alone does not have a statistically significant direct effect on economic growth (coefficient = 0.007, $p > 0.1$), likely due to infrastructure saturation. In contrast, inland cities still benefit directly from infrastructure improvements (coefficient = 0.014, significant at 5% level), which reflects their relatively underdeveloped transport networks and greater marginal returns from investment in roads.

Columns (3) and (4) present results for cities classified by administrative level: level 1 (provincial capitals and sub-provincial cities) and level 2–4 (lower-tier cities). In top-tier cities, the interaction term is large and significant (coefficient = 0.027), indicating that FDI and infrastructure are highly complementary. This is expected, as these cities have stronger institutions, better human capital, and more developed ecosystems to absorb and benefit from FDI. Even in lower-tier cities, the interaction term remains positive and statistically significant (coefficient = 0.008), though smaller in magnitude. This suggests that even less developed cities can benefit from complementarities between infrastructure and FDI, albeit to a lesser extent when basic conditions are in place.

Following China’s reform and opening-up, foreign-funded enterprises have been central to the country’s economic growth. To attract such investment, China introduced various preferential tax policies - most notably income tax reductions - which vary by region. [Demurger et al. \(2002\)](#) constructed a preferential policy index showing that cities within coastal open economic zones, special economic zones (SEZs), and economic and technological development zones consistently offer favorable tax rates to foreign investors. Since 2000, new zones such as the national synthetically reform testing districts, a second wave of SEZs, have emerged with similar objectives (see Table A.4). Additionally, bonded areas and tariff-free zones offer further incentives by exempting foreign goods from import duties. For this study, we classify cities affiliated with any of these zones as policy-supported, while cities not included are treated as lacking or having limited access to such preferential policies.

Columns (5) and (6) distinguish cities based on whether they are subject to preferential FDI policies, such as those located in special economic zones (SEZs), bonded zones, or national re-

form districts. Cities with preferential policies show a strong and significant complementary effect (interaction coefficient = 0.008), reinforcing the idea that targeted policy support amplifies the economic return from combining FDI and infrastructure investment. These cities may also benefit from streamlined administrative processes, tax incentives, and coordinated development planning that further boost FDI effectiveness. In non-policy cities, the interaction term is also positive and significant. This suggests that even without formal preferential treatment, road infrastructure and FDI can work together to support growth, although policy interventions clearly enhance the effect.

Notably, in Table A.5, the interaction between FDI and urban street infrastructure, $\ln(F) \times \ln(\text{UrbanStreet})$, is statistically significant only in the inland subsample (column 2), level-1 (top-tier) cities (column 3), and policy-supported cities (column 5). In contrast, the coefficients are small and imprecisely estimated for coastal cities (column 1), lower-tier (level 2–4) cities (column 4), and non-policy cities (column 6). This pattern suggests that within-city accessibility boosts the growth payoff from FDI where either (i) baseline intercity access is weaker (inland), (ii) administrative/absorptive capacity is high (top-tier), or (iii) supportive investment policies are in place (policy zones). Where these conditions are absent (coastal saturation, lower-tier capacity constraints, no policy support), urban street expansion does not measurably amplify the effect of FDI.

6 Discussion and Policy Implication

Our findings provide new evidence on the complementary relationship between foreign direct investment (FDI) and transportation infrastructure in driving regional economic growth. By extending the Solow growth model to incorporate both the direct productivity effects of road infrastructure and its moderating effect on the returns to FDI, we capture two important and distinct growth mechanisms. The empirical results consistently demonstrate that both FDI and road infrastructure individually contribute to long-term income growth, but more importantly, their interaction term exhibits a statistically significant and economically meaningful effect. This suggests that road infrastructure not only facilitates economic activity directly but also amplifies the efficiency gains from FDI.

This interaction mechanism has strong theoretical support. Roads reduce trade frictions, expand market access, and improve firm-level logistics, all of which make it easier for foreign firms to operate and for domestic firms to integrate into global value chains. The placement of road infrastructure in both the productivity function and the elasticity of FDI within our model reflects this dual role. Regions with better highway networks appear to gain disproportionately more from the presence of foreign capital, due to improved absorptive capacity and connectivity.

From a policy perspective, our results suggest that maximizing the growth impact of FDI requires complementary investments in transportation infrastructure. Regional governments seeking to attract and retain foreign capital should not focus solely on preferential tax policies or industrial parks. Instead, improving the physical connectivity between cities - particularly through the expansion and modernization of highways - can significantly raise the marginal returns to FDI. This is especially critical for third- and fourth-tier cities, which often receive lower levels of FDI due to limited accessibility and logistical bottlenecks.

Our findings also provide a rationale for coordinated national development strategies that integrate transport planning with industrial and investment policy. The observed synergy between infrastructure and FDI implies that fragmented, province-level efforts to attract foreign investment may be suboptimal unless matched with interregional infrastructure investments. The success of China's National Trunk Highway System and the clustering of economic activity around major corridors underscore the importance of such coordinated planning.

Furthermore, these dynamics have implications for regional inequality. Since infrastructure and FDI reinforce each other, their benefits may become concentrated in already well-connected urban hubs. To prevent increasing spatial inequality, policymakers should consider targeting road investment toward lagging or peripheral regions with untapped investment potential. Coupling this with incentives for FDI in those regions could foster more balanced development.

Finally, our paper contributes to the growing literature on endogenous growth by showing that physical infrastructure plays a more complex role than commonly assumed. It is not merely a background condition but a factor that can actively shift the growth trajectory when paired with external capital inflows. This reinforces the idea that infrastructure policy must be designed not just for capacity expansion but for strategic complementarity with broader economic goals.

7 Conclusion

China's economic transformation since 1978 has been underpinned by large-scale infrastructure investments and an openness to foreign capital. As the nation transitions from high-speed growth to high-quality development, policymakers have increasingly emphasized improving total factor productivity (TFP) and fostering more sustainable, innovation-driven economic dynamics. In this context, the role of continued infrastructure investment - especially in road networks - merits reexamination. Can further expansion of transport infrastructure still contribute meaningfully to economic efficiency and technological upgrading, particularly in a more mature growth phase?

This paper addresses that question by developing and testing an extended neoclassical growth framework that integrates foreign direct investment (FDI) and road infrastructure. Distinct from

existing studies, our model highlights two growth channels: (1) a direct effect of road infrastructure on productivity via reduced transport costs and market integration, and (2) an indirect effect, where roads enhance the productivity of FDI by improving foreign firms' operational reach and amplifying spillovers to domestic firms. Empirically, we find strong evidence that these two channels jointly contribute to long-run economic growth in China.

Using a province-city panel dataset spanning two decades, we estimate the effects of both FDI and road infrastructure and their interaction. Our baseline results confirm that both FDI and road infrastructure independently raise economic performance. More importantly, their interaction term exhibits a consistently positive and statistically significant effect on income growth, indicating that the marginal benefit of FDI is significantly higher in regions with more developed road networks. This supports the hypothesis that infrastructure enhances the absorptive capacity of host regions and enables foreign capital to be utilized more efficiently.

To address potential endogeneity in infrastructure and FDI placement, we construct and employ lagged and initial-value instrumental variables. The core findings remain robust across specifications, and the interaction effect becomes even more pronounced when instruments are applied. Further heterogeneity analysis suggests that the direct productivity effect of road infrastructure is more salient in inland and labor-intensive regions, where infrastructure constraints may still limit economic activity. In contrast, the FDI–infrastructure interaction effect is more uniformly distributed, pointing to its relevance even in more developed coastal provinces.

Taken together, the results reaffirm the strategic importance of infrastructure investment in China's evolving development model. While road expansion alone may yield diminishing returns in well-connected areas, its role as a productivity amplifier - particularly in combination with FDI - remains vital. Policymakers aiming to raise TFP and promote inclusive growth should therefore consider infrastructure investment not as a sunset policy tool, but as one that must be recalibrated to unlock synergies with capital inflows and regional innovation potential. Future investment strategies should prioritize integrated development plans that jointly coordinate transport infrastructure, foreign investment attraction, and local innovation ecosystems.

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Figures and Tables

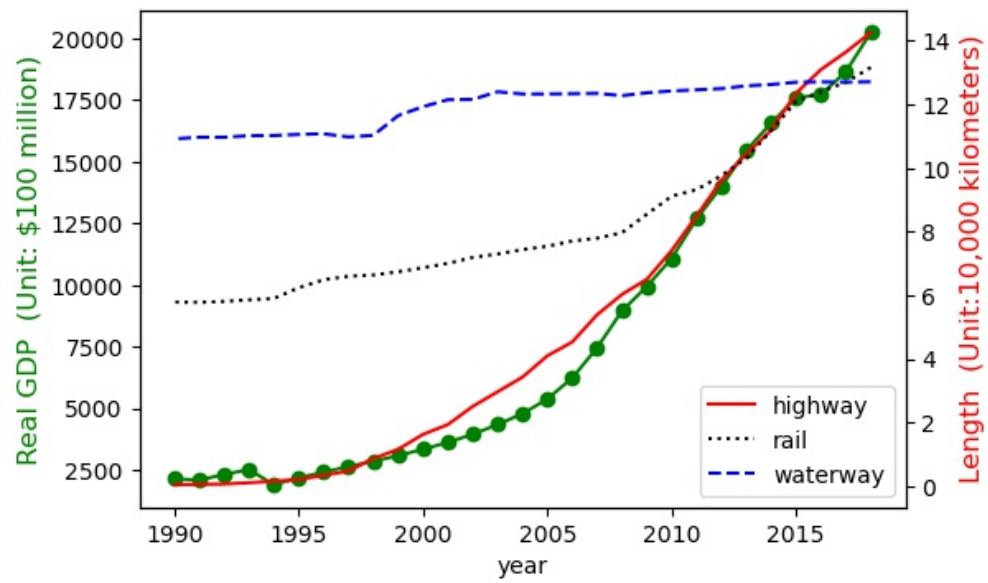


FIGURE 1 –TRANSPORTATION VS. REAL GDP

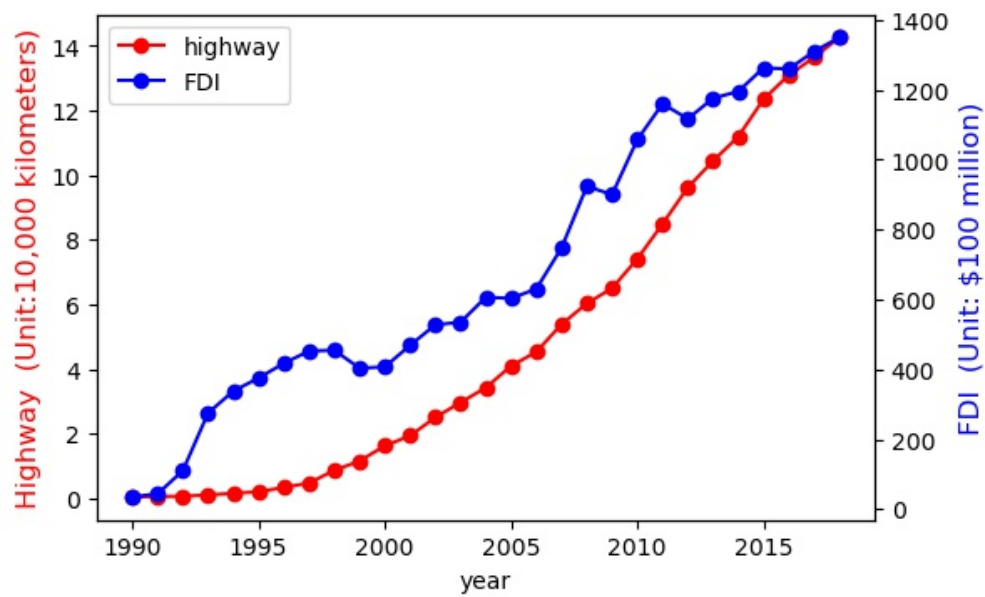


FIGURE 2 –HIGHWAY vs. FDI

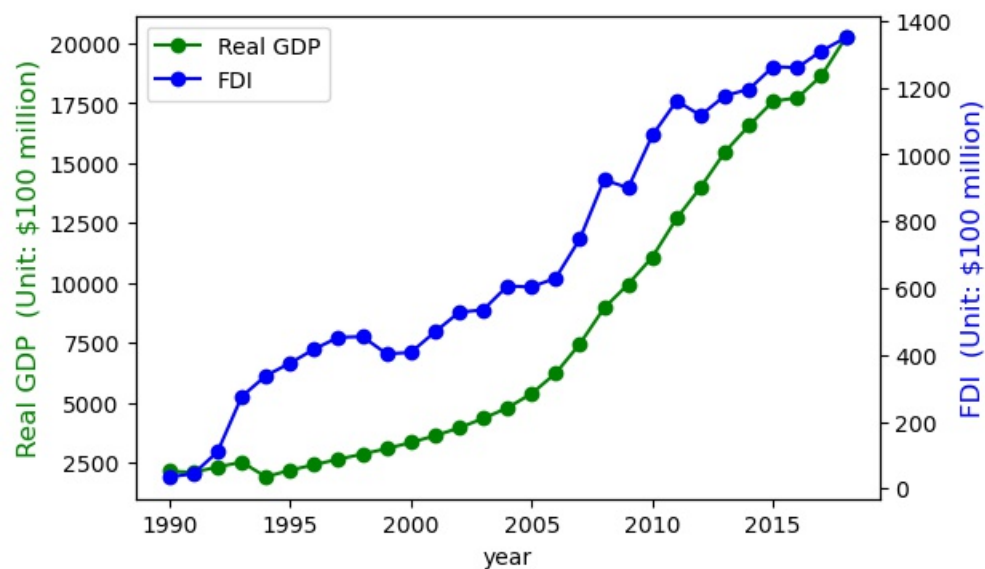


FIGURE 3 –REAL GDP vs. FDI

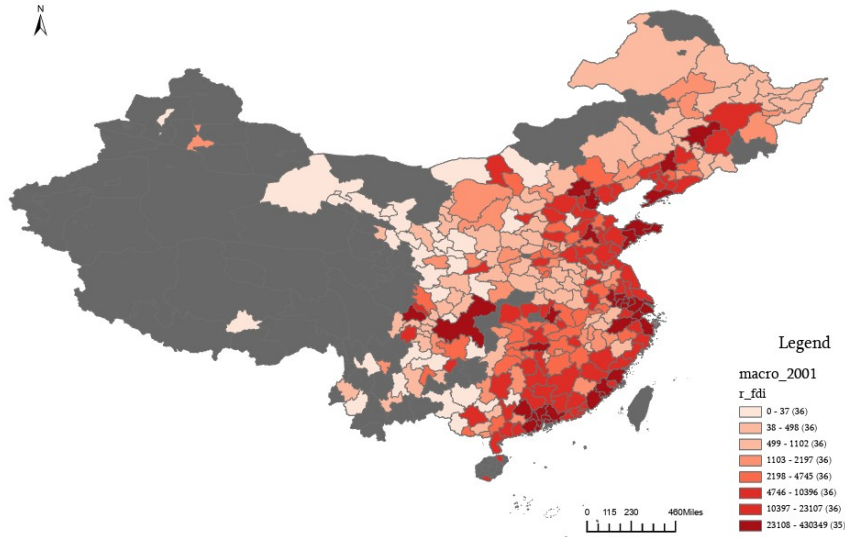


FIGURE 4 –FDI IN 2000

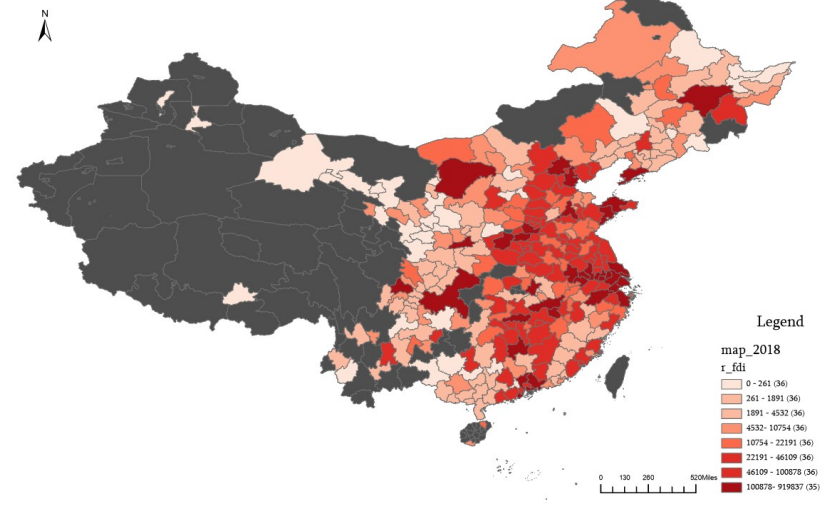


FIGURE 5 –FDI IN 2018

Notes: The two figures depict the new Foreign Direct Investment (FDI) inflow for the years 2000 and 2018. The classification method used for the FDI data is quantile classification, where each class contains an equal number of features. To illustrate, the cities shown in the darkest red color in the figures represent those with the highest levels of FDI for the respective years. The grey areas on the map are excluded from the sample cities, primarily consisting of autonomous regions and a few prefecture-level cities due to the significant amount of missing data.

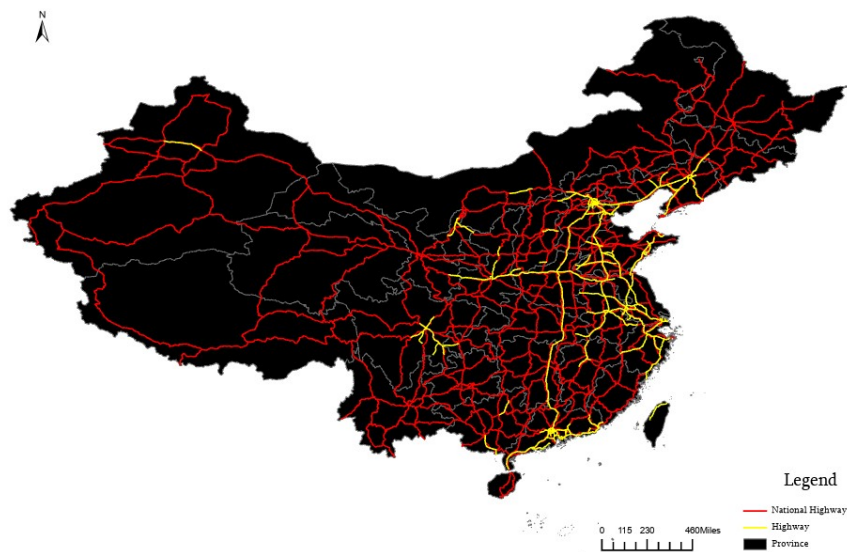


FIGURE 6 –ROAD IN 2000



FIGURE 7 –ROAD IN 2020

Notes: The two figures depict the national highway system(red line) and highway system(yellow line) of 2000 and 2020.

TABLE 1 –DESCRIPTIVE STATISTICS

	Mean	Std.Dev.	Obs
y, real GDP per capita (in Yuan)	21774.85	15807.99	4048
n, population growth rate	0.01	0.04	4048
s _k , ratio of investment to GDP	0.51	0.19	4048
s _h , share of population study in College	0.02	0.02	4048
F, stock of FDI (in 100,000,000 Yuan)	252.92	538.09	4048
Road, stock of road infrastructure (in 10,000 kilometre)	4.16	3.98	4048

TABLE 2 –CROSS-SECTION DATA REGRESSION RESULT

	Dependent Variable: logarithm difference of real GDP per capita				
	(1)	(2)	(3)	(4)	(5)
$\ln(y(0))$	-0.435*** (0.045)	-0.464*** (0.049)	-0.432*** (0.045)	-0.464*** (0.051)	-0.464*** (0.052)
$\ln(n + g + \delta)$	0.140 (0.130)	0.127 (0.130)	0.146 (0.131)	0.127 (0.132)	0.130 (0.138)
$\ln(s_k)$	0.217** (0.089)	0.217** (0.089)	0.216** (0.089)	0.217** (0.089)	0.217** (0.089)
$\ln(s_h)$	0.122*** (0.024)	0.113*** (0.025)	0.121*** (0.024)	0.113*** (0.025)	0.113*** (0.025)
$\ln(F)$		0.023 (0.017)		0.023 (0.018)	0.022 (0.022)
Road			0.003 (0.007)	-0.000 (0.007)	-0.001 (0.011)
$\ln(F) \times \ln(\text{Road})$					0.001 (0.011)
Constant	6.464*** (0.616)	6.536*** (0.617)	6.437*** (0.620)	6.537*** (0.624)	6.547*** (0.639)
Observations	220	220	220	220	220
R ²	0.354	0.359	0.354	0.359	0.359

Notes: This table reports the cross-sectional estimation using the average value of the entire period. The dependent variable is the log difference in GDP per capita between 1999 and 2018. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions.

TABLE 3 –PANEL DATA REGRESSION RESULT

	Dependent Variable: logarithm difference of real GDP per capita				
	(1)	(2)	(3)	(4)	(5)
$\ln(y(0))$	-0.246*** (0.019)	-0.264*** (0.022)	-0.279*** (0.023)	-0.296*** (0.026)	-0.317*** (0.027)
$\ln(n + g + \delta)$	-0.037 (0.033)	-0.030 (0.033)	-0.037 (0.030)	-0.031 (0.030)	-0.034 (0.029)
$\ln(s_k)$	0.150*** (0.022)	0.143*** (0.022)	0.144*** (0.021)	0.138*** (0.022)	0.131*** (0.021)
$\ln(s_h)$	0.118*** (0.020)	0.117*** (0.020)	0.112*** (0.019)	0.112*** (0.018)	0.096*** (0.017)
$\ln(F)$		0.027** (0.011)		0.025** (0.010)	0.022** (0.010)
Road			0.011** (0.005)	0.011** (0.005)	0.007* (0.004)
$\ln(F) \times \ln(\text{Road})$					0.010*** (0.003)
Constant	3.082*** (0.245)	3.158*** (0.246)	3.321*** (0.257)	3.389*** (0.260)	3.480*** (0.257)
City Fixed-effect	✓	✓	✓	✓	✓
Observations	870	870	870	870	870
R ²	0.400	0.407	0.421	0.427	0.437

Notes: This table reports the fixed-effect estimation under the four five-year periods. (1999-2003, 2004-2008, 2009-2013, 2014-2018) The dependent variable is the log difference of GDP per capita. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions.

TABLE 4 –ESTIMATION RESULTS BASED ON DIFFERENT VALUES DURING EACH PERIOD - INITIAL VALUES OF EACH 5-YEAR PERIOD

	Dependent Variable: logarithm difference of real GDP per capita				
	(1)	(2)	(3)	(4)	(5)
$\ln(y(0))$	-0.246*** (0.019)	-0.254*** (0.022)	-0.279*** (0.023)	-0.287*** (0.026)	-0.303*** (0.027)
$\ln(n + g + \delta)$	-0.037 (0.033)	-0.034 (0.033)	-0.037 (0.030)	-0.033 (0.030)	-0.035 (0.029)
$\ln(s_k)$	0.150*** (0.022)	0.150*** (0.022)	0.146*** (0.021)	0.145*** (0.022)	0.140*** (0.021)
$\ln(s_h)$	0.118*** (0.020)	0.118*** (0.020)	0.113*** (0.019)	0.113*** (0.018)	0.098*** (0.018)
$\ln(F)$		0.011 (0.011)		0.010 (0.010)	0.014 (0.011)
Road			0.010*** (0.004)	0.010** (0.004)	0.008** (0.004)
$\ln(F) \times \ln(\text{Road})$					0.005*** (0.002)
Constant	3.082*** (0.245)	3.130*** (0.255)	3.340*** (0.261)	3.384*** (0.273)	3.434*** (0.267)
City Fixed-effect	✓	✓	✓	✓	✓
Observations	870	870	870	870	870
R ²	0.400	0.401	0.419	0.420	0.428

Notes: The table reports the fixed-effect estimation using the initial values of road infrastructure and FDI stock instead of the average of each period. The dependent variable is the log difference in GDP per capita. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions.

TABLE 5 –PANEL DATA FOR SELECTED SUB-SAMPLES(RANK OF INNOVATION/PATENT) OF CITIES

	Innovation Index		Number of Patent	
	(1)	(2)	(3)	(4)
	Top Quarter	Bottom Quarter	Top Quarter	Bottom Quarter
$\ln(y(0))$	-0.329*** (0.054)	-0.213*** (0.050)	-0.319*** (0.054)	-0.258*** (0.051)
$\ln(n + g + \delta)$	-0.003 (0.037)	0.066 (0.042)	-0.043* (0.022)	0.044 (0.043)
$\ln(s_k)$	0.088** (0.036)	0.190*** (0.050)	0.060* (0.033)	0.210*** (0.045)
$\ln(s_h)$	0.047* (0.026)	0.025 (0.019)	0.064** (0.030)	0.034 (0.021)
$\ln(F)$	0.026 (0.023)	0.053*** (0.015)	0.050** (0.021)	0.022 (0.030)
Road	0.006 (0.007)	-0.034** (0.013)	0.009 (0.007)	-0.015 (0.015)
$\ln(F) \times \ln(\text{Road})$	0.017*** (0.005)	0.018 (0.018)	0.013** (0.006)	0.015 (0.020)
Constant	3.403*** (0.523)	2.575*** (0.513)	3.101*** (0.520)	3.015*** (0.537)
City Fixed-effect	✓	✓	✓	✓
Observations	311	78	287	113
R ²	0.373	0.652	0.396	0.544

Notes: This table reports the fixed-effect estimation under the four five-year periods and the selected sub-samples of cities. The dependent variable is the log difference of GDP per capita. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions. Because data on the innovation index or patents is unavailable for some cities, the total number of observations is inconsistent.

TABLE 6 –PANEL DATA FOR SELECTED SUB-SAMPLES(GEOGRAPHICAL/POLICY) OF CITIES

	Coastal Region		City Class		FDI Policy	
	(1) Coastal	(2) Inland	(3) level 1	(4) level 2-4	(5) w/	(6) w/o
$\ln(y(0))$	-0.332*** (0.049)	-0.333*** (0.033)	-0.380*** (0.091)	-0.335*** (0.029)	-0.223*** (0.040)	-0.371*** (0.031)
$\ln(n + g + \delta)$	0.049 (0.134)	-0.034 (0.028)	0.040 (0.059)	-0.042 (0.028)	0.052 (0.037)	-0.046 (0.029)
$\ln(s_k)$	0.117*** (0.029)	0.131*** (0.027)	0.023 (0.055)	0.139*** (0.024)	0.110*** (0.036)	0.132*** (0.025)
$\ln(s_h)$	0.060*** (0.017)	0.112*** (0.021)	0.018 (0.040)	0.097*** (0.018)	0.055*** (0.019)	0.110*** (0.021)
$\ln(F)$	0.003 (0.025)	0.022* (0.013)	0.025 (0.041)	0.021* (0.011)	0.020 (0.020)	0.029** (0.012)
Road	0.007 (0.006)	0.014** (0.006)	0.000 (0.010)	0.015*** (0.005)	-0.003 (0.006)	0.021*** (0.005)
$\ln(F) \times \ln(\text{Road})$	0.016*** (0.004)	0.007 (0.005)	0.027*** (0.009)	0.008* (0.004)	0.008* (0.005)	0.008* (0.004)
Constant	3.824*** (0.720)	3.683*** (0.308)	3.846*** (0.920)	3.626*** (0.276)	2.687*** (0.355)	3.955*** (0.312)
City Fixed-effect	✓	✓	✓	✓	✓	✓
Observations	202	664	108	758	209	657
R ²	0.417	0.462	0.393	0.459	0.369	0.491

Notes: This table reports the fixed-effect estimation under the four five-year periods and the selected sub-samples of cities. The dependent variable is the log difference of GDP per capita. The dependent variable is the log difference of GDP per capita. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions.

TABLE 7 –IV ESTIMATES - THE SECOND-STAGE RESULTS

	OLS	IV
	(1)	(2)
$\ln(y(0))$	-0.317*** (0.027)	-0.305*** (0.027)
$\ln(n + g + \delta)$	-0.034 (0.029)	-0.037 (0.029)
$\ln(s_k)$	0.131*** (0.021)	0.136*** (0.021)
$\ln(s_h)$	0.096*** (0.017)	0.098*** (0.018)
$\ln(F)$	0.022** (0.010)	0.010 (0.012)
Road	0.007* (0.004)	0.007* (0.004)
$\ln(F) \times \ln(\text{Road})$	0.010*** (0.003)	0.009*** (0.003)
Constant	3.480*** (0.257)	3.429*** (0.260)
City Fixed-effect	✓	✓
Observations	870	870

Notes: This table reports the second stage IV estimation. Column (1) is the benchmark regression(without IVs). All controls are not statistically significant. The dependent variable is the log difference in GDP per capita. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions.

Appendix A. Figures and Tables

TABLE A.1 –PANEL DATA REGRESSION RESULT

	w/o Urban Street	w/ Urban Street	
	(1)	(2)	(3)
$\ln(y(0))$	-0.317*** (0.027)	-0.328*** (0.033)	-0.340*** (0.034)
$\ln(n + g + \delta)$	-0.034 (0.029)	-0.027 (0.029)	-0.027 (0.030)
$\ln(s_k)$	0.131*** (0.021)	0.138*** (0.025)	0.133*** (0.025)
$\ln(s_h)$	0.096*** (0.017)	0.080*** (0.024)	0.079*** (0.025)
$\ln(F)$	0.022** (0.010)	0.006 (0.011)	0.009 (0.012)
Road	0.007* (0.004)	0.010* (0.005)	0.010* (0.005)
$\ln(F) \times \ln(\text{Road})$	0.010*** (0.003)	0.011** (0.005)	0.008* (0.005)
UrbanStreet		0.000 (0.000)	0.000 (0.000)
$\ln(F) \times \ln(\text{UrbanStreet})$			0.008** (0.004)
Constant	3.480*** (0.257)	3.599*** (0.331)	3.452*** (0.320)
City Fixed-effect	✓	✓	✓
Observations	870	724	720

Notes: This table reports the fixed-effect estimation under the four five-year periods. (1999-2003, 2004-2008, 2009-2013, 2014-2018) The dependent variable is the log difference of GDP per capita. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions.

TABLE A.2 –IV ESTIMATES - THE FIRST-STAGE AND REDUCED-FORM RESULTS

	Reduced-Form	First-stage Result		
	(1) ln(y)	(2) Road	(3) ln(F)	(4) ln(F) $\times$ ln(Road)
ln(y(0))	-0.303*** (.027)	-0.107** (0.0517)	0.003 (0.025)	0.250*** (0.076)
ln(n + g + δ)	-0.035 (0.029)	0.082** (0.037)	0.030 (0.022)	0.179** (0.070)
ln(s _k)	0.140*** (0.021)	.109*** (0.032)	0.202*** (0.029)	0.198*** (0.069)
ln(s _h)	0.098*** (0.018)	0.081** (0.032)	0.022 (0.028)	-0.147*** (0.046)
Road _{t-1}	0.008 ** (0.004)	0.951*** (0.014)	0.005 (0.003)	0.084*** (0.019)
ln(F) _{t-1}	0.014 (0.011)	0.009 (0.021)	0.857*** (0.018)	0.659*** (0.038)
ln(F) _{t-1} $\times$ ln(Road) _{t-1}	0.005*** (0.002)	0.001 (0.004)	-0.003 (0.003)	0.593*** (0.011)
Observations	870	870	870	870
First-stage F statistic	-	11409.49	891.88	4174.24

Notes: Column (1) reports the reduced form of the IV estimation. This table from column (2) to (3) reports the first stage of IV estimation. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions.

TABLE A.3 –PANEL DATA FOR SELECTED SUB-SAMPLES(RANK OF INNOVATION/PATENT) OF CITIES

	Innovation Index		Number of Patent	
	(1)	(2)	(3)	(4)
	Top Quarter	Bottom Quarter	Top Quarter	Bottom Quarter
$\ln(y(0))$	-0.389*** (0.065)	-0.129 (0.084)	-0.359*** (0.067)	-0.243*** (0.067)
$\ln(n + g + \delta)$	-0.005 (0.036)	0.004 (0.038)	-0.032 (0.024)	0.032 (0.039)
$\ln(s_k)$	0.120*** (0.035)	0.164*** (0.047)	0.094** (0.039)	0.192*** (0.055)
$\ln(s_h)$	0.009 (0.028)	-0.021 (0.025)	-0.002 (0.028)	0.028 (0.038)
$\ln(F)$	0.031 (0.025)	0.013 (0.033)	0.054** (0.025)	0.029 (0.058)
Road	0.008 (0.008)	-0.072*** (0.019)	0.005 (0.008)	-0.014 (0.022)
$\ln(F) \times \ln(\text{Road})$	0.017*** (0.006)	0.038 (0.025)	0.017*** (0.006)	-0.003 (0.044)
UrbanStreet	0.000 (0.000)	-0.000* (0.000)	0.000 (0.000)	-0.000 (0.000)
$\ln(F) \times \ln(\text{UrbanStreet})$	0.011** (0.005)	0.030 (0.021)	0.010** (0.004)	-0.006 (0.031)
Constant	3.302*** (0.569)	1.219 (0.725)	2.768*** (0.537)	2.928*** (0.753)
City Fixed-effect	✓	✓	✓	✓
Observations	260	69	239	94
R ²	0.467	0.703	0.505	0.669

Notes: This table reports the fixed-effect estimation under the four five-year periods and the selected sub-samples of cities. The dependent variable is the log difference of GDP per capita. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions. Because data on the innovation index or patents is unavailable for some cities, the total number of observations is inconsistent.

TABLE A.4 –CHINA’S REGIONAL PREFERENTIAL POLICIES

Types	Cities
Special Economic Zones	1988/04 All cities in Hainan Province
	1989/04 Shenzhen
	1994/03 Xiamen
	1996/03 Zhuhai
	1996/03 Shantou
The National Synthetically Reform Testing District	2005/6 Shanghai
	2006/5 Tianjin
	2007/6 Chongqing, Chengdu
	2007/12 Wuhan, Changsha, Xiangtan
	2009/5 Shenzhen
	2010/04 Shenyang
	2010/12 All cities in Shanxi Province
	2011/03 Yiwu
	2012/08 Dongxing, Ruili, Manzhouli
	2013/05 Nantong
Bonded Area	2013/06 All cities in Heilongjiang Province
	2014/09 Shantou
	1987 Shenzhen(Shatou)
	1990 Shanghai(Gaoqiao)
	1991 Tianjin, Shenzhen(Futian)
	1992 Haikou, Qingdao, Yantai,Dalian
	1993 Shantou
	1996 Shenzhen(Yantian)
	2002 Ningbo
	2006 Suzhou
	2008 Qinzhou, Pingxiang
	2009 Tianzhu, Suifenhe
	2010 Chengdu, Chongqing, Zhengzhou
	2011 Xian, All cities in Xinjiang Province, Wuhan, Weifang
	2012 Taiyuan, Chifeng, Zhoushan
Coastal Open Economic Zones	2014 Nanyang, Ganzhou
	2018 Weihai
	1984 Dalian, Qinhuangdao, Shanghai, Nantong, Tianjin, Ningbo, Guangzhou, Zhanjiang, Yantai, Fuzhou, Lianyungang, Qingdao

TABLE A.5 –PANEL DATA FOR SELECTED SUB-SAMPLES(GEOGRAPHICAL/POLICY) OF CITIES

	Coastal Region		City Class		FDI Policy	
	(1)	(2)	(3)	(4)	(5)	(6)
	Coastal	Inland	level 1	level 2-4	w/	w/o
$\ln(y(0))$	-0.390*** (0.069)	-0.333*** (0.041)	-0.462*** (0.114)	-0.332*** (0.037)	-0.253*** (0.047)	-0.371*** (0.043)
$\ln(n + g + \delta)$	-0.013 (0.208)	-0.024 (0.031)	0.043 (0.087)	-0.032 (0.031)	0.049 (0.038)	-0.045 (0.033)
$\ln(s_k)$	0.101*** (0.037)	0.146*** (0.032)	0.065 (0.065)	0.144*** (0.029)	0.109** (0.046)	0.142*** (0.031)
$\ln(s_h)$	0.062* (0.034)	0.089*** (0.031)	-0.046 (0.046)	0.083*** (0.026)	0.082 (0.053)	0.073*** (0.028)
$\ln(F)$	-0.002 (0.038)	0.009 (0.015)	0.039 (0.049)	0.005 (0.013)	0.013 (0.022)	0.005 (0.015)
Road	-0.001 (0.012)	0.011* (0.006)	-0.002 (0.015)	0.010* (0.006)	-0.000 (0.009)	0.011* (0.006)
$\ln(F) \times \ln(\text{Road})$	0.020*** (0.007)	0.006 (0.008)	0.028** (0.011)	0.008 (0.007)	0.003 (0.007)	0.014** (0.007)
UrbanStreet	0.000 (0.000)	0.000 (0.000)	0.000 (0.000)	-0.000 (0.000)	-0.000 (0.000)	0.000 (0.000)
$\ln(F) \times \ln(\text{UrbanStreet})$	0.012 (0.007)	0.008** (0.004)	0.019** (0.009)	0.006 (0.004)	0.014* (0.007)	0.007 (0.004)
Constant	3.776*** (0.913)	3.447*** (0.403)	3.255*** (0.929)	3.484*** (0.348)	2.652*** (0.426)	3.705*** (0.401)
City Fixed-effect	✓	✓	✓	✓	✓	✓
Observations	170	550	93	627	179	541
R ²	0.475	0.511	0.501	0.504	0.410	0.537

Notes: This table reports the fixed-effect estimation under the four five-year periods and the selected sub-samples of cities. The dependent variable is the log difference of GDP per capita. The dependent variable is the log difference of GDP per capita. Robust standard errors clustered at the city level are reported in parentheses. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$. Refer to Table (1) for variable definitions.